

## Current and future issues of active noise control

Jiri Tichy

*Graduate Program in Acoustics, The Pennsylvania State University,  
Applied Science Building, University Park, PA 16802, USA*

Active control of sound fields is an emerging technology which has a great potential to become superior to many passive low frequency noise control methods. It is based on fundamental physical principles which include acoustical coupling of sources, destructive wave interference, and control of enclosure modes. The implementation of active noise control requires placing secondary sources and microphones into the controlled field. Most systems use adaptive filters to control the secondary sources to achieve maximum noise reduction. The engineering applications will require developing optimization techniques for secondary sources and controlling microphone locations. Also new types of robust actuators and controlling sensors will be needed. A brief summary of existing and future applications is provided.

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### 1. INTRODUCTION

The concept of controlling and possibly reducing the sound field of a source, using interfering waves from another source with  $180^\circ$  phase shift, can be traced back to Lord Rayleigh. Although the principle of active field control is simple, the complexity of reducing the acoustic energy in the entire sound field, or of reducing the sound power radiation from a complex source, can be enormous. Therefore, it should be no surprise that it took more than one hundred years to convert Rayleigh's idea into practical applications. The evolution of active sound field control is marked by two milestones. In 1933 Paul Lueg patented a scheme for reducing the sound propagation in a tube. About twenty years later, Harry Olson,<sup>1)</sup> a brilliant inventor of acoustic transducers, constructed the first hearing protection device by sensing the incident noise wave and re-radiating it with reversed phase so that the sound pressure is minimized at the ear.

None of these attempts at active noise reduction were too successful because all the necessary conditions for a destructive wave interference and source

coupling were not well understood. To achieve substantial noise reduction, the secondary wave has to match the amplitude and phase (except for  $180^\circ$  shift) of the primary wave with high precision. Progress in computational techniques in acoustics, advancements in control theory, invention of adaptive filtering and availability of fast digital electronics at affordable prices, provided both the knowledge and hardware to achieve promising progress in implementing this new technology.

### 2. PRINCIPLES OF ACTIVE CONTROL

Implementation of active noise control is based on two distinct tasks: determination of an optimum location of the cancellation sources, and time control of the amplitude and phase of the radiated sound waves. In addition, there are severe requirements on the operation of the mechanical and electronic parts of the system. Transducers for field sensing and control must sustain the operation under adverse environmental conditions. In essence, the secondary sources must be capable of radiating acoustic power equal to the power of the primary sources. The controlling configurations and elec-

tronics need to respond to changes in both the noise sources and their environment, maintain an overall operational stability and have an affordable cost.

Although the basic principles of active noise control are simple, extensive research and development during the 1980's resulted in relatively few commercial applications. In spite of this slow progress, we can expect growing industrial interest, particularly in low frequency applications for which noise reduction, using traditional techniques, is not feasible.

Results achieved in the 1980's, in both research and development of active attenuators, have shown that one of the major tasks to be solved is the understanding and further exploration of acoustical principles and source configurations that are useable for noise reduction implementation.

Most active noise control applications can be classified into two categories: the reduction of sound power radiation at the source, and the sound pressure reduction in the field. None of the strategies developed for these approaches have any useful value unless a suitable criterion for noise reduction, often called cost function, has been formulated in such a manner as to become an implementable component of the controlling part of the overall noise control system. The lack of suitable acoustical controlling schemes is the main reason that practical active controls have been achieved in only the simplest source configurations or sound propagation paths.

## 2.1 Control by Acoustical Coupling

The main physical principle of source radiation reduction is based on acoustical coupling with other sources that confines a substantial portion of the acoustical volume flow into the near field of both the primary and secondary sources so that only a reduced portion of acoustic power propagates into the far field. Often cited examples of this mechanism include the conversion of a monopole source into a dipole, quadrupole, or a higher order source configuration.<sup>2,3)</sup> It should be noted, for the sake of completeness, that any one of these single source configurations does not result in minimization of the sound power radiation of the primary-secondary source configuration. Depending on the primary source configuration, a combination of the monopole, dipole and higher order sources is needed to minimize the total radiated power.

To illustrate this mechanism of sound power reduction and the engineering problems related to the design of the systems, the fundamental aspects of sound power reduction will be shown. Let's consider first a harmonically oscillating primary source of volume velocity  $Q_p$  and a secondary source located at a distance  $d$ . The total power radiated by both sources is minimized if the secondary source strength  $Q_s$  is

$$Q_s = -Q_p [\sin kd/kd] \quad (1)$$

the sound power radiated by the primary sources is

$$P_p = \frac{k^2 qc}{8\pi} Q_p^2 \left[ 1 - \left( \frac{\sin kd}{kd} \right)^2 \right] \quad (2)$$

while the secondary source power  $P_s = 0$ .

Equation (2) reveals that a substantial reduction of radiated power can be achieved only for  $kd \ll 1$  so that the primary-secondary source distance must be less than one-half wavelength. As stated above, the secondary source does not radiate any power. However, it has to be capable of generating volume flow  $Q_s$  (Eq. (1)) which, in essence, is equal to the volume flow of the primary source. This may put a severe requirement on the source construction and continuous operation.

It is interesting to show that the sound power of a dipole source  $P_D$  is always greater than the power of an optimized source pair. The power ratio is given by

$$\frac{P_p + P_s}{P_D} = \frac{1 + \frac{\sin kd}{kd}}{2} < 1 \quad (3)$$

where  $P_D$  is the total power radiated by a dipole with the individual source strengths  $\pm Q_p$ . Equation (2) also reveals that whatever distance both sources may have, the phase shift between their surface vibration is  $180^\circ$ .

The situation involving the sound power reduction by a more complex source will be illustrated on a model consisting of  $N$  primary point sources located at  $x_p$  with the strengths represented by a complex vector  $Q_p$  and  $M$  secondary sources at  $x_s$  with the strength vector  $Q_s$  (Fig. 1). The interaction between the sources can be characterized by the mutual impedance defined from

$$p_{ij} = Z_{ij} Q_j \quad (4)$$

where the  $p_{ij}$  is the pressure at the location of the  $i$ -th source generated by the  $j$ -th source. It can be shown that the total power radiated by all sources

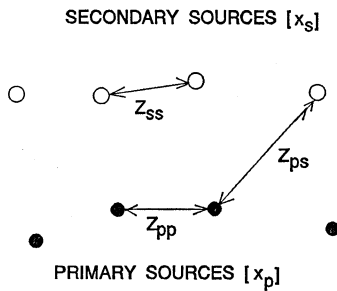


Fig. 1 Primary and Secondary Sources.

is minimized if the secondary source vector  $Q_s$  is adjusted using

$$Q_s = -D^{-1}B \quad (5)$$

where the matrix  $D = 1/2 \operatorname{Re}[Z_s(X_s)]$  and  $B = 1/2 \operatorname{Re}[Z_p(X_s)]Q_p$ ;  $Z_s(X_s)$  is a matrix of mutual impedance between the secondary sources and  $Z_p(X_s)$  is a matrix between the primary and secondary sources.

The total minimum radiated power for the secondary sources optimized by Eq. (5) is

$$P_0 = C - B^H A^{-1} B \quad (6)$$

where  $C = 1/2 Q_p^H \operatorname{Re}[Z_p(X_p)]Q_p$  is a scalar representing the power radiated by the primary sources in the absence of the secondary sources. Superscript H represents the Hermitian transpose of the matrix.

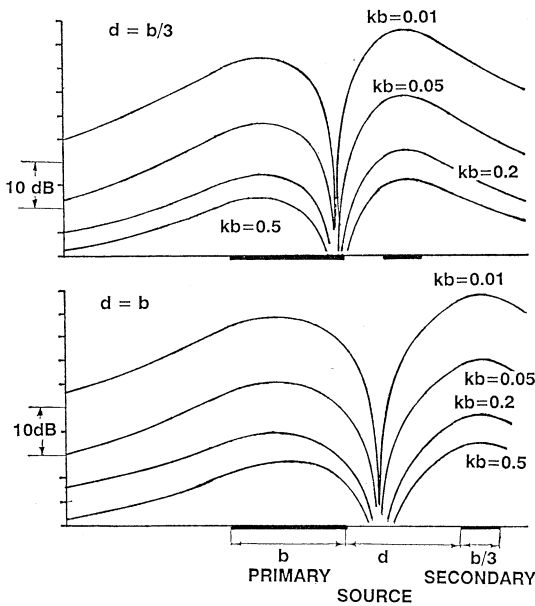
As stated earlier, the total power of the source configuration can, in essence, be minimized only if the sources are located closer than one-half wavelength. The optimization process (Eq. (2)) of the secondary source results in a substantial decrease of the sound power radiation from this source after the system reaches steady state. However, the sources have to be capable of generating the volume velocity that matches the volume velocity of the primary source and, as discussed later, the availability of low frequency secondary sources with such capabilities may be limited.

Let's now turn to the engineering aspects of sound power radiation reduction at the source. Very few actual sources can be characterized as pure monopoles. The principal representatives of monopole source approximations are terminations of ducts, exhaust pipes, power plant and other silencers, provided that their cross-dimensions are sufficiently small.<sup>4)</sup> Small vibrating machines or structural elements can often be treated as dipole radiators.

An extension of the theoretical approach to minimize the sound power output from discrete primary and secondary monopole sources can result in determining the optimum strengths of the secondary sources. However, the theory will not provide specific information of the optimum size and locations of the secondary sources except to point out that they have to be located as close to the primary sources as possible.

An inseparable component of the engineering solution is the need to automatically control the secondary sources to minimize the overall sound power radiation. Due to both the changes of the primary source sound output, and to the environmental changes affecting the sound propagation speed and the wavefront shape, the choice of the controlling criterion and its implementability by the electronics is crucial for the system performance. Most of the electronic controls are based on adaptive filtering which adjusts the secondary source power output until a minimum voltage is achieved at a chosen reference point. According to this scheme, a pressure sensor is placed in a suitably chosen field point and the adaptive electronic controls adjust the input into the auxiliary sources for the minimum sound pressure at the reference point. The crucial part of optimizing the performance of the system is the selection of the suitable location at which the sound pressure will be minimized. The pressure sensor has to be placed at a location where the sound pressure will have a minimum after the secondary sources are adjusted for the total minimum sound power output. Such a location can be found either experimentally or by computational techniques such as the finite surface element technique.

The implications of the zero control concept, and the difficulty of selecting an optimum secondary source location, can be illustrated by an example of a monopole conversion into a dipole. The dipole sound pressure minimum is in the symmetry plane of the two sources and the controlling sensor has to be placed in this plane. The modeling of actual sources as dipoles is approximate, and its validity is limited to the prediction of power reduction from the distance of the "acoustical centers" between the primary and secondary sources. An excellent example on the sound pressure minima was presented by Kido<sup>5,6)</sup> who studied the near field of area sources in order to establish an insight into the



**Fig. 2** Relative sound pressure level distribution along a line  $b/3$  above two sources adjusted for maximum sound power reduction (Kido<sup>6)</sup>).

optimum source configuration and the sensitivity of an exact sensor location on the optimum control. Figure 2 shows one example from his work on pressure distribution from two rectangular pistons with the side lengths  $b$  and  $b/3$ . Their volume velocities were adjusted for minimum total power radiation. The pressure was calculated along a line parallel to the plane of sources at the height  $b/3$ . We can see that the pressure minima are not symmetrically located between the sources and that, depending on  $kb$ , they can be very sharp which makes the microphone location for an optimum control very critical. If the sources are complex, the exact location of the error sensors has to be determined either experimentally or predicted by finite surface element computer techniques. The optimum location of the error sensor will be further affected by the time, and variability of the particle velocity distribution over the sources.

The idea of using the sound pressure as an acoustical control quantity comes from the early noise control work on ducts, where the plane wave propagating energy is linked in a simple manner to the potential energy density and thus conveniently

sensed by a pressure microphone. However, when sensing the pressure minimum in open space, serious adverse effects on the pressure minimum can occur due to reflections from the environment. As mentioned earlier, the adaptive filters adjust the secondary source sound output to sound pressure minimum which consists of the sum of all contributing pressures, including the reflections. As a phaser diagram quickly reveals, the effect of the reflections can result in misadjustments of the secondary sources with respect to the minimum total power output.<sup>7)</sup> The reflections, which for discrete frequencies are coherent with the primary sound, can be very strong. For instance, if the noise source is controlled in a room, strong low frequency standing waves can develop and affect the noise cancellation function.

The ultimate goal of any active noise control device is to reduce the state variables of the sound field, such as the sound pressure, in the far field of the primary sources. The far field pressure depends on the radiated sound power and, therefore, the next generation of controllers should be based on a power related quantity. In addition, the near field of the primary-secondary source combination is marked by small size minima and large size maxima, and the control based on pressure minimum is too sensitive to spatial stability of source radiation. On the other hand, sound intensity-based control, with multiple input signal processing, will avoid this disadvantage.

Most of the sound sources are complex. The vibrating surfaces can be represented as an assembly of monopole sources which vibrate with different phases. Although radiation reduction based on the coupling principle can be applied, a great deal of research will be needed to explore achievable theoretical limits and to provide design engineers with sufficient guidelines for the secondary source configurations suitable for specific types of primary sources. In addition, the radiation pattern of such a multisource configuration is complex, and a single control based on minimizing the sound pressure at a single field point will have to be replaced by a multipoint and multichannel control. Similar implications linked to source variability and the need to develop different types of controls, which were discussed earlier for monopole sources, also applies here.

### 3. CONTROL OF RADIATED SOUND FIELD

Practical attempts to decrease the source radiation into the free field, based on a different principle than coupling, were attempted with relatively good success. As shown by Jessel,<sup>8)</sup> the sound field of any source can, in theory, be confined within the interior volume of a surface circumscribing the source and completely canceled everywhere outside this surface. As follows from the Helmholtz radiation integral and the Huygens principle, this field modification can be achieved by a suitable continuous distribution of monopole and dipole sources on the circumscribing surface (Fig. 3). An engineering solution, based on constructing continuously variable combinations of monopole and dipole sources, is very difficult, and so far has not been attempted. However, Angevine<sup>9)</sup> explored the practical possibility of using a discrete, rather than a continuous distribution of monopoles and dipoles located on a hemisphere circumscribing an electric power transformer. Each source was controlled independently for the minimum pressure outside the hemisphere. Not surprisingly, the greater the number of sources, the higher the attenuation. The concept of the secondary source envelope around the primary source seems to be rather complex, but progress in designing new types of sound sources, and progress in a multichannel control which is currently pursued for artificial intelligence purposes, will undoubtedly also be utilized for active noise control.

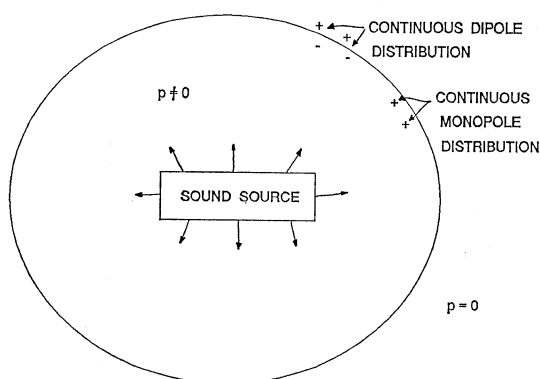


Fig. 3 Outer zone of complete sound field suppression by a continuous distribution of monopole and dipole sources.

### 4. FIELD CONTROL BY INTERFERENCE

It has been shown that global control of radiation from sources can be achieved only if the primary and secondary sources are located in close proximity. If the secondary source is several wavelength distant from the primary source, their interference pattern can be adjusted by the controlling microphone so that a minimum sound pressure is achieved

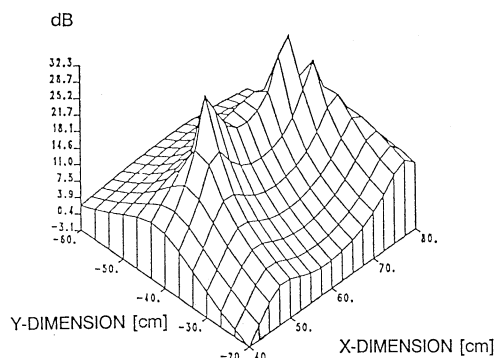


Fig. 4 Sound pressure level reduction achieved by a 2 channel system in the vicinity of two controlling microphones. The microphones are positioned at the peak level reduction.

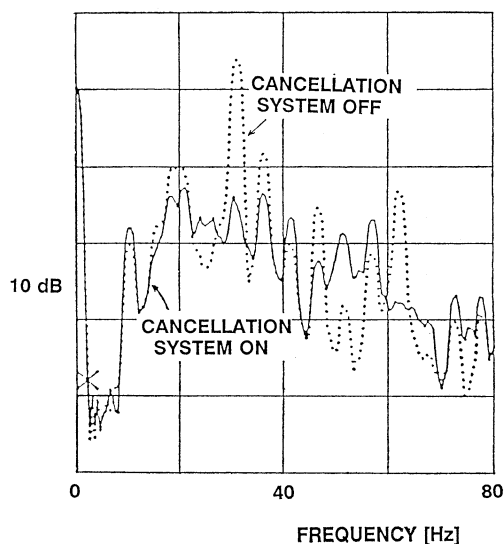


Fig. 5 Low frequency noise reduction at the driver's seat of an automobile.

at a desired field location. However, the size of the quiet zone is approximately only about one-tenth of the wavelength and therefore the engineering applications are limited to low frequencies. In addition, the overall energy in the sound field is increased by the energy radiated from the secondary source. To minimize this energy increase, the secondary source should be placed close to the quiet zone which will thus be in its near field and the energy radiated by this secondary source will be kept at minimum.

## **5. MODAL CONTROL IN ENCLOSURES**

Another principle of active field control is linked to sound fields in enclosures with the goal of achieving an overall energy minimization. Global energy reduction can be realistically achieved at low frequencies with light sound field damping so that the modal overlap is small and a narrow band signal will excite only a relatively few modes. To achieve global field reduction, the primary source mode excitation has to be counteracted by a set of secondary sources which have properly adjusted complex volume velocities.<sup>10)</sup> The practical limitations of this technique depend on the modal damping which determines the number of "substantially" excited primary modes. Although the volume velocity vector of secondary sources is unique for their specific location, a sufficient number of secondary sources is needed to achieve satisfactory engineering results. Figure 4 shows local sound level reduction using a two channel system. Realistic solutions are limited to cases of low modal density. As a consequence, only relatively small enclosures, such as automobiles and similar cabins, (Fig. 5) can be treated for periodically excited low frequency noise.

This brief summary of the acoustics of active sound field reduction control shows that the configuration concerning the primary and secondary source distribution, and the ways the field minimization can be acoustically controlled, are complex. All these have to be solved prior to engaging any electronic hardware. Progress in electronic control systems has been so rapid during the 1980's that the electronics-related tasks of noise reduction technology have become substantially easier to accomplish than the design of acoustical configurations.

## **6. CONTROL ELECTRONICS**

Most of the acoustic schemes of active field reduc-

tion require continuous minimization of the selected acoustical or vibrational field descriptors transduced into electrical voltages. The controlling signal is derived from noise sources, and after filtering which adjusts its proper magnitude and phase, this signal is reradiated by the secondary sources. To achieve any noise reduction at all, the sound field at the controlling location has to be coherent with the noise source because the controller will not affect any portion of the noise field which is not represented in its input. Also, the controller which models the transfer function, characterizing the propagation path from the primary and secondary sources to the controlling sensor, has to follow any changes of the acoustical properties of this path. This can be achieved by using adaptive filtering which, in its practical implementation, is most often based on the LMS algorithm.<sup>11)</sup>

Because of needed flexibility, it became very popular to implement adaptive filtering by programming fast Digital Signal Processing boards hosted by a PC computer. DSP boards are capable of up to 50 megaflops per second. This allows the design of digital filters with many coefficients, and a high signal sampling rate. It also permits the implementation of other tasks such as propagation path identification, which may become an inseparable part of the signal processing for active noise and vibration control.

## **7. TRANSDUCERS**

Although the success of system implementation depends on transducers, very little progress has been achieved in developing specialized transducers capable of continuous, reliable operation under all kinds of environmental conditions. This is particularly true for the auxiliary noise sources.

As stated earlier, the auxiliary sources must be as capable of radiating equal volume velocity as the noise sources. The principal applications of active control technique are at low frequencies. To supply the active system with sufficient power, the generated volume flow must increase with decreasing frequency.

### **7.1 Loudspeakers**

All existing airborne active noise control installations use conventional loudspeakers which were originally designed for music reproduction. The efficiency of these speakers is inherently low, and

their mechanical design is based on the need to continuously supply relatively low average music power with only occasional high power peaks.

To increase the volume velocity in a narrow frequency band, a variety of Helmholtz resonator configurations can be used.<sup>12)</sup> At the resonances, the excursion of the speaker cone is usually minimized while the resonator provides peak volume velocity output. However, the diaphragm of the driver has to be capable of handling high forces required to drive the resonator. The high forces and excursions of the loudspeaker diaphragm heavily strain both the cone and suspension materials and, therefore, loudspeakers commercially available for music systems are generally not suitable. By using the resonators, the sound radiation can be concentrated as needed into a narrow frequency band. An excellent narrow band sound source is obtained by placing a Helmholtz resonator in front of a speaker in a closed box baffle. The bandwidth, as well as the efficiency, can be controlled by the volume of the Helmholtz resonator. If two ports are desirable or useable, a system with two resonators, one in front and one over the back of the driver, provides good design flexibility while keeping the loudspeaker excursion low. In most cases, the use of bass-reflex design is unsuitable. An alternative to the Helmholtz resonator type of enclosures is a folded tube with a one to three ratio of the tube lengths in the front and the rear of the speaker. This configuration is particularly suitable for very low and narrow band frequency applications which also require the actual speaker to be sheltered and placed as far away as possible from the sound injection area.

## 7.2 Microphones

Most experimental set-ups use condenser microphones designed for sound recording. This kind of microphone might be usable in room-like environments such as automobiles or aircraft cabins. However, industrial or outdoor environments require a robust design. A good example of a mechanically suitable transducer is the piezoelectric pressure sensors, as long as their sensitivity, frequency and phase response, low response to vibrations and other environmental effects, such as temperature and humidity, are adequate.

In conclusion, most active noise control applications will require substantial improvements in transducer design, the utilization of new materials

and modifications of the existing mechanical constructions.

## 8. INDUSTRIAL APPLICATION FOR ACTIVE NOISE CONTROL

Most of the work on active noise control systems in the 1980's occurred in laboratories and was funded by institutions rather than by industry. In spite of an initial strong wave of interest, industry has mostly waited on the sidelines and watched for decisive proof of the success of this new technology. The role of researchers in the 1990's will be to assist industry with the development, engineering, and installation of the active control systems.

Active noise control techniques can be labeled as one of the high technologies of noise control. It cannot be implemented by a cut-and-try approach which is often the mode many practitioners choose, in lieu of a systematic approach to design procedures based on an understanding of the principles and information obtained from research.

In spite of many difficulties, there are application areas in which the active noise systems have been successfully implemented. Historically, development started with the control of one-dimensional wave propagation in ducts because of its relative simplicity and the need to attenuate low frequency noise.<sup>7,13)</sup> Attenuation of low frequency noise by means of passive silencers is either practically impossible for lack of space or prohibitively expensive. Active systems were developed which can substantially reduce both the periodic and random noise in air conditioning and other ducts and pipes at frequencies lower than the first cross-mode cut-off.<sup>7,14)</sup> Installations have been completed to reduce the low frequency noise of power plant exhaust silencers.<sup>15)</sup> Work is in progress on silencers for diesel engines and automobile mufflers.<sup>16)</sup> Another area of successful application is the noise reduction in small enclosed spaces such as vehicles and similar size cabins and aircraft interiors.<sup>17,18)</sup> We will soon see active noise reduction of periodic noise components on small fans and climate systems.<sup>19)</sup> Work is also in progress to actively reduce the tonal components on large size industrial fans. And Harry Olson would certainly be pleased to see that active control is successfully applied on headsets and ear defenders which otherwise offer minimal transmission loss at low frequencies.

The future of active noise control is bright and

will undoubtedly reflect the results of the research of the past decade.

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